Section 11
INTRO TO SE

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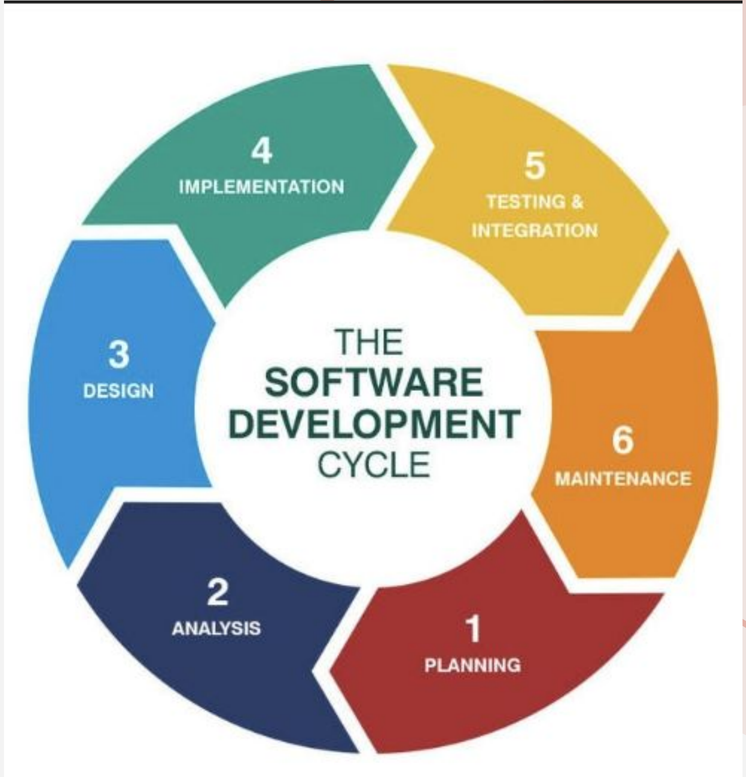
Developing Product

Software Development Life Cycle (SDLC):

Software development life cycle (SDLC) is a structured process that is used to design, develop, and test good-quality software. SDLC, or software development life cycle, is a methodology that defines the entire procedure of software development step-by-step.



Software Development Life Cycle (SDLC)

- Stage-1: Planning and Requirement Analysis
- Stage-2: Defining Requirements
- Stage-3: Designing Architecture
- Stage-4: Developing Product
- Stage-5: Product Testing and Integration
- Stage-6: Deployment and Maintenance of Products

What is Integration Testing?

- After building separate modules, we **combine them and test them together**.

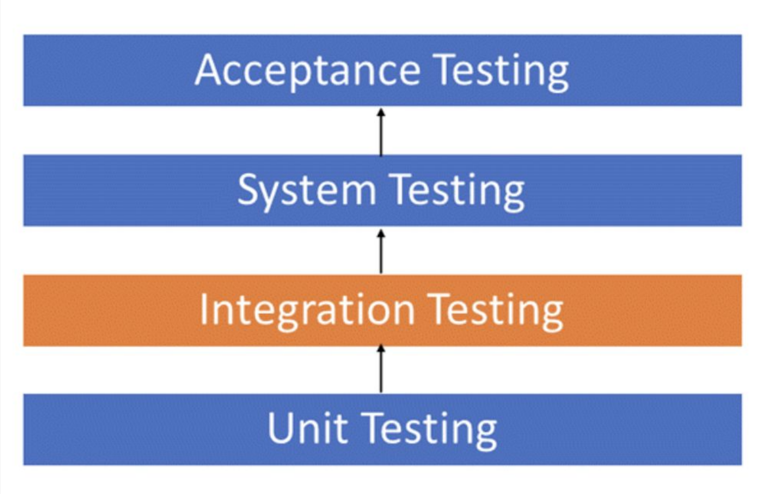
Because:

- Different developers build different parts
- Each part works alone
- But may fail when connected

Checking **communication between modules**
Making sure **data flows correctly**

3. Why do Integration Testing?

First test small parts
Then test combined parts
Then whole system



Reasons we need integration testing:

1. Different developers → different logic
2. Requirements may change after coding
3. Module connections may fail
4. Database interaction errors
5. Hardware/software interface issues
6. Poor error handling

In short:
Even if parts work alone, they might break together

Types of Integration Testing

Big Bang Approach :

- Incremental Approach:

which is further divided into the following

- Top Down Approach
- Bottom Up Approach
- Sandwich Approach – Combination of Top Down and Bottom Up

Big Bang Testing

- is an Integration testing approach in which all the components or modules are integrated together at once and then tested as a unit. This combined set of components is considered as an entity while testing. If all of the components in the unit are not completed, the integration process will not execute.
- Advantages:

  Convenient for small systems.
- Disadvantages:

  Fault Localization is difficult.
- Given the sheer number of interfaces that need to be tested in this approach, some interfaces link to be tested could be missed easily.

- Since the Integration testing can commence only after "all" the modules are designed, the testing team will have less time for execution in the testing phase.

- Since all modules are tested at once, high-risk critical modules are not isolated and tested on priority. Peripheral modules which deal with user interfaces are also not isolated and tested on priority.

Incremental Testing

Idea:

- Combine modules **step by step**
- Test each step

Much safer and more controlled

In the Incremental Testing approach, testing is done by integrating two or more modules that are logically related to each other and then tested for proper functioning of the application.

Then the other related modules are integrated incrementally and the process continues until all the logically related modules are integrated and tested successfully.

- Incremental Approach, in turn, is carried out by two different Methods:
- Bottom Up
- Top Down

Bottom up testing

Start with **low-level modules first**
Move upward

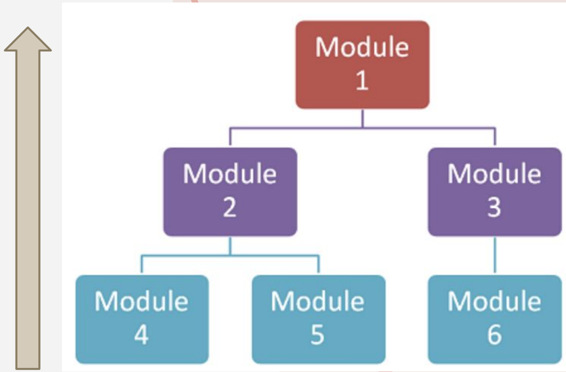
Test small components → combine → go higher

Advantages:

- Fault localization is easier.
- No time is wasted waiting for all modules to be developed unlike Bigbang approach

Disadvantages:

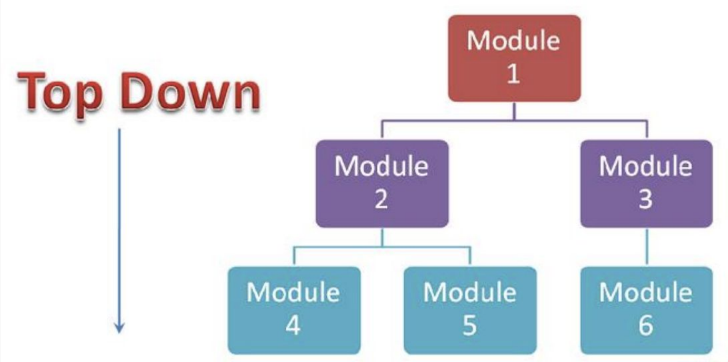
- Critical modules (at the top level of software architecture) which control the flow of application are tested last and may be prone to defects.
- An early prototype is not possible



Top down testing

- Start with **top-level modules**
- Move downward
- following the control flow of software system.
- The higher level modules are tested first and then lower level modules are tested and integrated in order to check the software functionality.
- Stubs are used for testing if some modules are not ready.

Advantages:
• Fault Localization is easier.
• Possibility to obtain an early prototype.
• Critical Modules are tested on priority; major design flaws could be found and fixed first.
Disadvantages:
• Needs many Stubs.
• Modules at a lower level are tested inadequately.

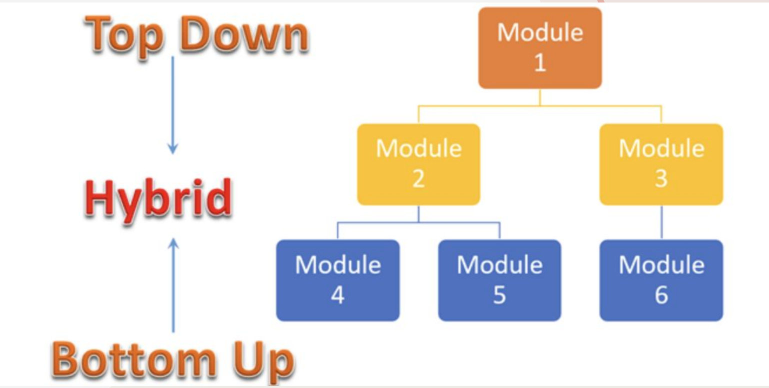


Sandwich Testing

a strategy in which top level modules are tested with lower level modules at the same time lower modules are integrated with top modules and tested as a system.

It is a combination of Top-down and Bottom-up approaches therefore it is called Hybrid Integration Testing.

It makes use of both stubs as well as drivers.



Integration Testing Tools

Two categories:

1. Language-specific tools

- Pytest → Python
- jQuery → JavaScript/PHP
- H2 → Java

2. Multi-language tools

- Selenium
- Cucumber
- VectorCast

Used to automate testing